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Candidate Number

BIOLOGY PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5 and 7.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) Present your answers in paragraphs wherever appropriate.
- (7) The diagrams in this section are **NOT** necessarily drawn to scale.
- (8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.



SECTION B

Answer **ALL** questions. Put your answers in the spaces provided.

1. For each type of the blood cells listed in column 1, select from column 2 **one** phrase that correctly describes its function. Put the appropriate letter in the space provided. (3 marks)

Column 1

Lymphocytes _____

Blood platelets _____

Red blood cells _____

Column 2

A. Involved in blood clotting

B. Involved in oxygen transport

C. Involved in antibody production

D. Involved in phagocytosis

E. Involved in transporting hormones

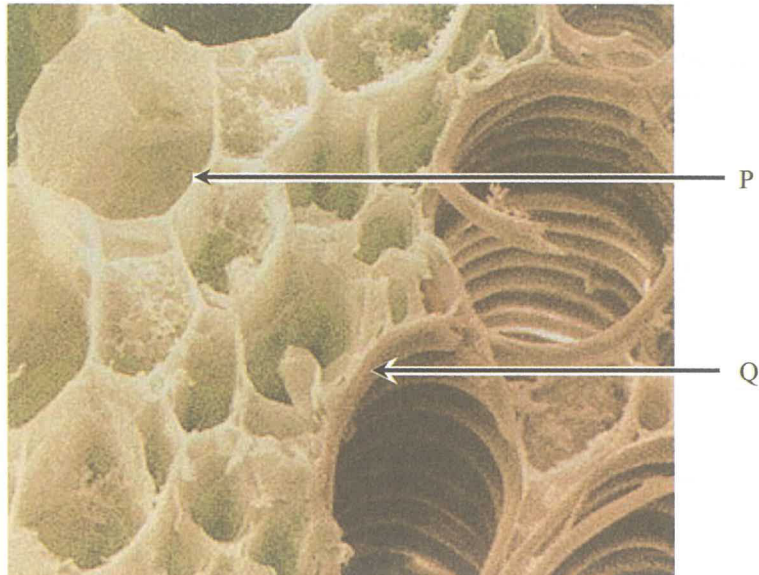
2. Dengue Fever is a vector borne disease transmitted by mosquitoes. The table below shows two methods adopted by the Government to break the transmission link of this disease. Complete the table below to show how each method works and comment on its advantage **or** disadvantage. (4 marks)

<i>Method</i>	<i>How it works</i>	<i>Comment (advantage or disadvantage)</i>
Spraying of pesticides or larvicidal oil around mosquito's habitat		
Clearance of accumulated water in a neighbourhood		

Answers written in the margins will not be marked.

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3. The electron micrograph below shows part of the stem of a plant containing two cell types, P and Q:



- (a) Based on the photograph shown, state the difference between cell types P and Q in the structure indicated by the arrow heads. (1 mark)

- (b) Describe how these cells contribute to the support of the plant. (4 marks)

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4. The following key can be used for identifying organisms under the same phylum:

1a	Absence of eyes	-----	2
1b	A pair of eyes	-----	3
2a	Six legs	-----	Class A
2b	More than six legs	-----	Class B
3a	Six legs	-----	Class C
3b	More than six legs	-----	Class D

(a) Using the above key, identify which class organism X shown in the photograph below belongs to. (1 mark)



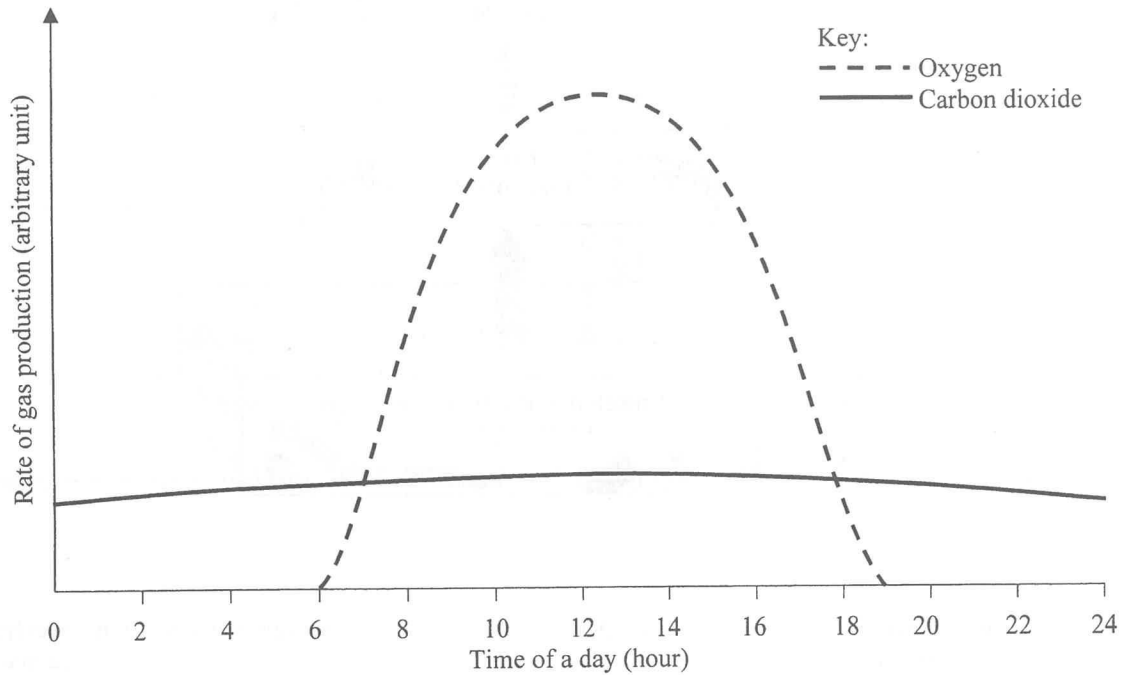
(b) Suggest a characteristic of the habitat of organism X. Explain your answer. (2 marks)

(c) A newly found organism Y has a pair of eyes and fewer than six legs. Although it is believed that this organism belongs to this phylum, it cannot be identified by using the above key. Explain why this problem occurs. (1 mark)

(d) Suggest **one** way to collect more information which can be used for deciding whether organism Y belongs to this phylum. (2 marks)

Answers written in the margins will not be marked.

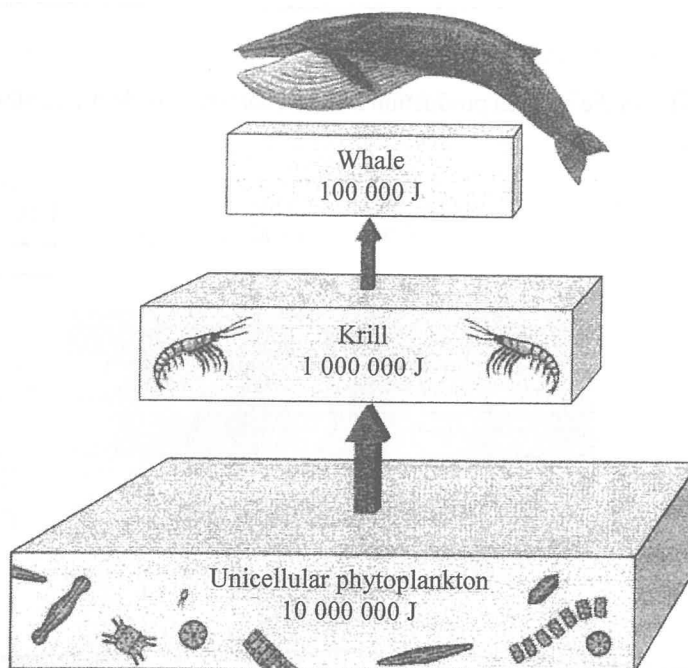
5. The graph below shows the oxygen production rate and carbon dioxide production rate of a local plant on a summer day:



- (a) State the times at which there is no net exchange of gases into or out of the leaves. (1 mark)
- (b) Sketch a line on the above graph to show the oxygen production rate of the plant on a winter day. (2 marks)
- (c) The area below the line showing the oxygen production rate is usually greater than the area below the line showing the carbon dioxide production rate. Explain the importance of this observation. (4 marks)

Answers written in the margins will not be marked.

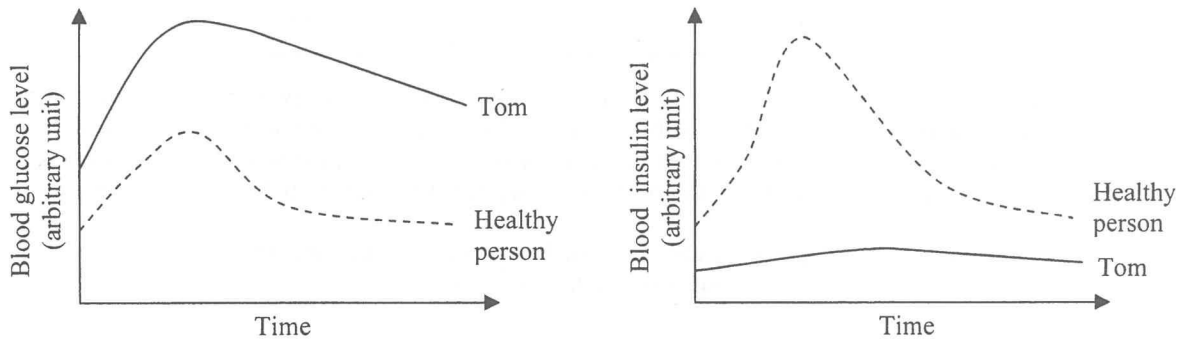
6. The diagram below shows a food chain in an ocean. The energy content of each trophic level is shown in the boxes:



- (a) In the space below, calculate the percentage decrease in energy content from unicellular phytoplankton to krill. (2 marks)
- (b) Give *two* reasons why there is a decrease in energy content from a lower trophic level to a higher one. (2 marks)
- (c) In another food chain, the producer is tree and the primary consumer is caterpillar. Would the percentage decrease in energy content be greater or smaller than the value found in (a)? Explain your answer. (2 marks)

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7. Tom suffers from diabetes. His doctor asked him to drink, after overnight fasting, a large volume of a glucose solution. After that, blood samples were taken at regular time intervals, to measure insulin and glucose contents. The following graphs show the changes in Tom's blood glucose level and blood insulin level after the test, and those of a healthy person:



- (a) Which type of diabetes does Tom suffer from? Explain your answer. (4 marks)

- (b) Explain the difference in blood glucose response to the oral consumption of glucose solution between Tom and the healthy person. (3 marks)

- (c) What medical treatment should Tom be given? (1 mark)

Answers written in the margins will not be marked.

8. The photograph below shows the appearances of some kernels of a corn:

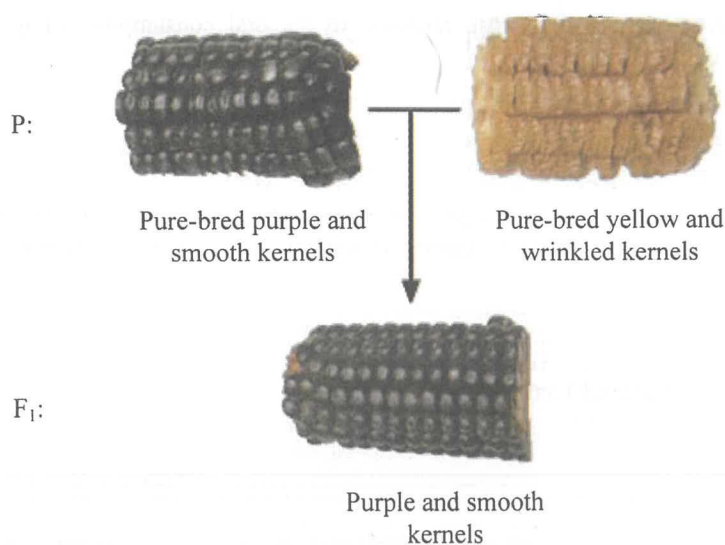


Kernel	Appearance
A	Purple and smooth
B	Purple and wrinkled
C	Yellow and smooth
D	Yellow and wrinkled

The purple colour is produced by a pigmented layer within the kernels. If the layer is not pigmented, the yellow colour of the inner tissue becomes visible. Whether the kernel is smooth or wrinkled is due to the type of food stored inside it. Smooth kernels (starchy corn) store starch while wrinkled kernels (sweet corn) store soluble sugars. The surface of the sweet corns becomes wrinkled when the corn dries up.

- (a) With reference to osmosis, explain why the kernels of sweet corn become wrinkled when they dry up but the kernels of starchy corn remain smooth. (4 marks)

- (b) The two traits of the kernels are controlled by genes located on different homologous chromosomes. The following diagram shows the result of a cross between two pure-bred corn plants, one with purple and smooth kernels and the other with yellow and wrinkled kernels:

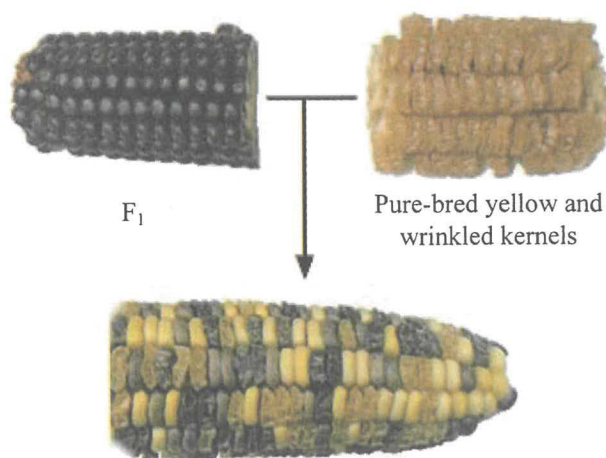


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- (i) Based on the results of the cross, deduce which phenotypes are dominant. (3 marks)

- (ii) The F_1 generation produced was crossed with pure-bred corn plants with yellow and wrinkled kernels, as shown below:



Explain the results of the cross using Mendel's law of inheritance. (4 marks)
(Marks will not be awarded for genetic diagrams.)

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- (c) When Mendel proposed how traits are inherited, chromosomes had not yet been discovered. In your opinion, how did Mendel come up with his hypothesis? (3 marks)

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9. Drugs X and Y may inhibit enzymes involved in glycolysis, the Krebs cycle or oxidative phosphorylation. To study the effects of the drugs, some muscle cells were isolated and treated with these two drugs separately in the presence of oxygen. The cellular levels of ATP, NADH, and pyruvate were determined. The results are shown in the table below:

	ATP	NADH	Pyruvate
Control (without treatment)	100%	100%	100%
Drug X	2%	3%	5%
Drug Y	20%	15%	150%

The data for the control are set as 100% for comparative purpose.

- (a) Suggest the key process that is inhibited by drug X. Explain your answer. (3 marks)

- (b) Suggest the key process that is inhibited by drug Y. Explain why there is an accumulation of pyruvate in the muscle cells after treatment with drug Y. (3 marks)

- (c) Instead of incubating in the presence of oxygen, the untreated muscle cells were incubated under anaerobic conditions. Predict the change in the cellular ATP, NADH and lactate levels. (3 marks)

- (d) A student would like to study the enzymes involved in glycolysis, the Krebs cycle and oxidative phosphorylation separately. Suggest which cellular components he needs to isolate for the investigation. (3 marks)

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10. In mammals, the production of bile salts is very limited at birth and during early developmental stages. In an investigation of the effect of bile supplementation on fat digestion in piglets, four groups of piglets were fed with the same diet except that bile was added to diets in the proportion of 0, 0.15%, 0.30% and 0.45% respectively for 15 days. Faecal samples were collected each day for analysis and the body weights of the piglets were monitored. The results are shown in the table below:

	Bile level in diet (%)			
	0	0.15	0.30	0.45
Average initial body weight (kg)	4.90	4.46	4.65	4.52
Average final body weight (kg)	7.15	7.36	7.81	7.92
Average dry weight of fat in the faeces (%)	6.52	5.81	4.65	3.78

- (a) How do bile salts help with fat digestion? (2 marks)

- (b) What was the effect of bile supplementation on fat digestion in the above investigation? Explain your answer. (2 marks)

- (c) Suggest why it is important to monitor the body weight of the piglets in this investigation. (1 mark)

- (d) Suggest an *in vitro* experimental method that allows a direct measurement of the digestion of fat in a laboratory. (3 marks)

For the following question, candidates are required to present their answer in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Mitosis and meiosis are important processes that ensure the continuity of life. Contrast the two processes and state the significance of their differences. (11 marks)

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A large rectangular box with a thin black border, containing 25 horizontal lines for writing. The lines are evenly spaced and extend across the width of the box.

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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